10 A small mass on a vibrating platform undergoes simple harmonic motion. The total energy of the mass is 18 mJ.

The amplitude of the oscillation is now doubled, and the time period is increased to be three times as long.

What is the new total energy of the mass?

A 8 mJ **B** 12 mJ **C** 36 mJ **D** 72 mJ

Ans: A

total energy of a simple harmonic oscillator,

$$E_{total} = \frac{1}{2}mv_0^2 = \frac{1}{2}m(\omega x_0)^2 \propto \left(\frac{x_0}{T}\right)^2 \Rightarrow \frac{E_{new}}{E_{old}} = \left(\frac{2}{3}\right)^2 = \frac{4}{9} \Rightarrow E_{new} = \frac{4}{9}E_{old} = \frac{4}{9}(18\text{ mJ}) = 8\text{ mJ}$$

11 The diagram below shows three polarisers W, X and Y.